Functional Analysis

Tutorial 3

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1 Assignments

Problem 1.1. Prove that for ℓ^p spaces, (e_k) defined by

$$e_k = (0, 0, \ldots, 1, 0, \ldots) \text{ (1 in the k-th position)}$$

is a Schauder basis.

Proof. Let $1 \leq p < \infty$ and $x = (x_1, x_2, \ldots) \in \ell^p$. Define the partial sums

$$s_n = \sum_{k=1}^{n} x_k e_k = (x_1, x_2, \ldots, x_n, 0, 0, \ldots).$$

Then

$$\|x - s_n\|_p^p = \sum_{k=n+1}^{\infty} |x_k|^p \to 0 \quad \text{as } n \to \infty,$$

since the tail of a convergent series tends to zero. Hence $s_n \to x$ in ℓ^p-norm. This shows that every $x \in \ell^p$ can be represented as

$$x = \sum_{k=1}^{\infty} x_k e_k,$$

with convergence in ℓ^p. The uniqueness of the coefficients is obvious: if $\sum_{k=1}^{\infty} \alpha_k e_k = 0$, then each $\alpha_k = 0$. Therefore (e_k) is a Schauder basis for ℓ^p. $\square$

Problem 1.2. Let X and Y be linear spaces. Show that $X \cap Y$ is a linear space, while $X \cup Y$ is not necessarily a linear space.

Proof. First, we show $X \cap Y$ is a linear space. Since X and Y are linear spaces, they both contain the zero vector, so $0 \in X \cap Y$. Let $u, v \in X \cap Y$ and α be a scalar. Then $u, v \in X$ and $u, v \in Y$. Since X is linear, $u + v \in X$ and $\alpha u \in X$. Similarly, $u + v \in Y$ and $\alpha u \in Y$. Hence $u + v \in X \cap Y$ and $\alpha u \in X \cap Y$. Thus $X \cap Y$ is closed under addition and scalar multiplication, so it is a linear space.

Second, we show $X \cup Y$ need not be a linear space. Consider $X = \{(x, 0) : x \in \mathbb{R}\}$ and $Y = \{(0, y) : y \in \mathbb{R}\}$ in $\mathbb{R}^2$. Both are linear subspaces. Their union $X \cup Y$ consists of the coordinate axes. Take $u = (1, 0) \in X$ and $v = (0, 1) \in Y$. Then $u + v = (1, 1)$, which is not in $X \cup Y$. Thus $X \cup Y$ is not closed under addition, hence not a linear space. $\square$

Problem 1.3. In ℓ^∞, let Y be the set of all sequences with only finitely many non-zero terms. Show that Y is a subspace of ℓ^∞, but Y is not closed in ℓ^∞.

Proof. Clearly $Y \subset \ell^\infty$. The zero sequence is in Y. If $x, y \in Y$, then each has only finitely many non-zero terms, so $x + y$ also has only finitely many non-zero terms, hence $x + y \in Y$. Similarly, for any scalar α, $\alpha x \in Y$. Thus Y is a subspace of ℓ^∞.

To show Y is not closed, consider the sequence $(x^{(n)})$ in Y defined by

$$x^{(n)} = \left(1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{n}, 0, 0, \ldots\right).$$

Each $x^{(n)}$ has only n non-zero terms. In ℓ^∞, we have

$$\|x^{(n)} - x\|_\infty \to 0 \quad \text{where } x = \left(1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{k}, \ldots\right).$$

Indeed, $\|x^{(n)} - x\|_\infty = \sup_{k>n} 1/k = 1/(n+1) \to 0$. But x has infinitely many non-zero terms, so $x \notin Y$. Thus Y does not contain all its limit points, hence Y is not closed. $\square$

Problem 1.4. Prove that the existence of a Schauder basis in a normed space X implies the separability of X.

Proof. Let $(e_k)_{k=1}^\infty$ be a Schauder basis for X. Then every $x \in X$ has a unique representation $x = \sum_{k=1}^\infty \alpha_k e_k$ with convergence in norm. Consider the set

$$D = \left\{ \sum_{k=1}^n q_k e_k : n \in \mathbb{N}, q_k \in \mathbb{Q} \text{ (or } \mathbb{Q} + i\mathbb{Q} \text{ if the field is complex)} \right\}.$$

D is countable because it is a countable union of countable sets. We show D is dense in X. Given $x = \sum_{k=1}^\infty \alpha_k e_k \in X$ and $\varepsilon > 0$, choose n such that $\| \sum_{k=1}^\infty \alpha_k e_k - \sum_{k=1}^n \alpha_k e_k \| < \varepsilon/2$. Then for each $k = 1, \ldots, n$, choose a

rational (or complex rational) q_k such that $|\alpha_k - q_k| < \varepsilon/(2nM)$ where $M = \max_{1\le k\le n} \|e_k\|$. Then

$$\left\|\sum_{k=1}^{n} \alpha_k e_k - \sum_{k=1}^{n} q_k e_k\right\| \le \sum_{k=1}^{n} |\alpha_k - q_k|\|e_k\| < n \cdot \frac{\varepsilon}{2nM} \cdot M = \varepsilon/2.$$

Hence $\|x - \sum_{k=1}^{n} q_k e_k\| < \varepsilon$. Thus D is dense in X, so X is separable. □

Problem 1.5. Let K be a compact metric space and S is a closed subset of K. Show that S is compact.

Proof. We prove using sequential compactness. Since K is a compact metric space, it is sequentially compact: every sequence in K has a convergent subsequence. Let (x_n) be a sequence in S. Since $S \subset K$, (x_n) is also a sequence in K. By sequential compactness of K, there exists a subsequence (x_{n_k}) converging to some point $x \in K$. But S is closed, and (x_{n_k}) is in S, so its limit x must belong to S. Thus every sequence in S has a convergent subsequence with limit in S. Hence S is sequentially compact, and since S is a metric space (as a subset of a metric space), it is compact.

Alternatively, using open covers: let $\{U_i\}_{i\in I}$ be an open cover of S. Since S is closed, $K \setminus S$ is open. Then $\{U_i\}_{i\in I} \cup \{K \setminus S\}$ is an open cover of K. By compactness of K, there exists a finite subcover of K. Removing $K \setminus S$ if necessary, we obtain a finite subcollection of $\{U_i\}$ that covers S. Hence S is compact. □

Problem 1.6. If $\dim X < \infty$, show that one can choose $\alpha = 1$ in F.Riesz's lemma.

Proof. Recall F. Riesz's lemma: Let X be a normed space, $Y \subset X$ a proper closed subspace, and $0 < \alpha < 1$. Then there exists $x \in X$ with $\|x\| = 1$ such that $\operatorname{dist}(x, Y) \ge \alpha$.

When $\dim X < \infty$, we can strengthen α to 1. Let Y be a proper closed subspace (in finite dimensions, every subspace is closed). Consider the quotient space X/Y with the quotient norm $\|x+Y\| = \inf_{y\in Y} \|x+y\|$. Since $\dim X/Y = \dim X - \dim Y$ is finite, the unit sphere in X/Y is compact. Hence there exists an element $x + Y \in X/Y$ with $\|x + Y\| = 1$ and the infimum is attained: there exists $y_0 \in Y$ such that $\|x + y_0\| = 1$. Let $z = x + y_0$, then $\|z\| = 1$ and

$$\operatorname{dist}(z, Y) = \inf_{y\in Y} \|z + y\| = \inf_{y\in Y} \|x + y_0 + y\| = \inf_{y'\in Y} \|x + y'\| = \|x + Y\| = 1.$$

Thus we have found $z \in X$ with $\|z\| = 1$ and $\operatorname{dist}(z, Y) = 1$. So we can take $\alpha = 1$. □

2 Expansions

We now present one of the most fundamental representation theorems in functional analysis: the identification of the dual space of ℓ^p for $1 < p < \infty$.

Proposition 2.1 (Dual of ℓ^p)**.** Let $1 < p < \infty$ and let q be the conjugate exponent, i.e. $\frac{1}{p} + \frac{1}{q} = 1$. Define the map $\Phi : \ell^q \to (\ell^p)^*$ by

$$\Phi(y)(x) = \sum_{n=1}^{\infty} x_n y_n, \qquad \text{for } y = (y_n) \in \ell^q,\ x = (x_n) \in \ell^p.$$

Then Φ is an isometric isomorphism. Consequently, $(\ell^p)^* \cong \ell^q$.

Proof. We break the proof into four steps.
Step 1: Φ is well-defined and bounded. Let $y \in \ell^q$ and $x \in \ell^p$. By Hölder's inequality,

$$|\Phi(y)(x)| = \left| \sum_{n=1}^{\infty} x_n y_n \right| \leq \sum_{n=1}^{\infty} |x_n||y_n| \leq \|x\|_p \|y\|_q.$$

Thus the series converges absolutely, and $\Phi(y)$ is a linear functional on ℓ^p satisfying

$$\|\Phi(y)\|_{(\ell^p)^*} = \sup_{\|x\|_p = 1} |\Phi(y)(x)| \leq \|y\|_q. \tag{1}$$

Step 2: Φ is an isometry. We show the reverse inequality $\|\Phi(y)\| \geq \|y\|_q$. If $y = 0$ the claim is trivial. Assume $y \neq 0$. For each $N \in \mathbb{N}$ define a vector $x^{(N)} \in \ell^p$ by

$$x_n^{(N)} = \begin{cases} \operatorname{sgn}(y_n)\, |y_n|^{q/p}, & \text{if } 1 \leq n \leq N, \\ 0, & \text{if } n > N. \end{cases}$$

Then

$$\|x^{(N)}\|_p = \left(\sum_{n=1}^{N} |y_n|^q \right)^{1/p}.$$

Moreover,

$$\Phi(y)(x^{(N)}) = \sum_{n=1}^{N} |y_n|^{q/p} \operatorname{sgn}(y_n)\, y_n = \sum_{n=1}^{N} |y_n|^{q/p+1} = \sum_{n=1}^{N} |y_n|^q = \|x^{(N)}\|_p^p.$$

Hence,

$$\frac{|\Phi(y)(x^{(N)})|}{\|x^{(N)}\|_p} = \frac{\|x^{(N)}\|_p^p}{\|x^{(N)}\|_p} = \|x^{(N)}\|_p^{p-1} = \left(\sum_{n=1}^{N} |y_n|^q \right)^{1/q}.$$

Since $\|\Phi(y)\| \geq \frac{|\Phi(y)(x^{(N)})|}{\|x^{(N)}\|_p}$ for every N, letting $N \to \infty$ gives

$$\|\Phi(y)\| \geq \|y\|_q. \tag{2}$$

Combining (1) and (2) we obtain $\|\Phi(y)\| = \|y\|_q$; therefore Φ is an isometry.
Step 3: Φ is linear. Linearity of Φ follows directly from the linearity of the sum: for $y, z \in \ell^q$ and $\alpha \in \mathbb{C}$,

$$\Phi(\alpha y+z)(x) = \sum_{n=1}^{\infty} x_n(\alpha y_n+z_n) = \alpha \sum_{n=1}^{\infty} x_n y_n + \sum_{n=1}^{\infty} x_n z_n = \alpha\Phi(y)(x)+\Phi(z)(x).$$

Step 4: Φ is surjective. Let $f \in (\ell^p)^*$. Define a sequence $y = (y_n)$ by $y_n = f(e_n)$, where e_n is the n-th standard unit vector (i.e. $e_n(k) = \delta_{nk}$). We claim that $y \in \ell^q$ and that $f = \Phi(y)$.

First, for each N construct the same auxiliary vector $x^{(N)}$ as in Step 2:

$$x_n^{(N)} = \begin{cases} \operatorname{sgn}(y_n)\,|y_n|^{q/p}, & 1 \le n \le N, \\ 0, & n > N. \end{cases}$$

Then, as before, $\|x^{(N)}\|_p = \left(\sum_{n=1}^{N} |y_n|^q\right)^{1/p}$ and

$$f(x^{(N)}) = \sum_{n=1}^{N} x_n^{(N)}\, y_n = \sum_{n=1}^{N} |y_n|^q.$$

Since $|f(x^{(N)})| \le \|f\|\|x^{(N)}\|_p$, we have

$$\sum_{n=1}^{N} |y_n|^q \le \|f\| \left(\sum_{n=1}^{N} |y_n|^q\right)^{1/p},$$

which implies

$$\left(\sum_{n=1}^{N} |y_n|^q\right)^{1/q} \le \|f\|.$$

Letting $N \to \infty$ gives $\|y\|_q \le \|f\|$; hence $y \in \ell^q$.

Finally, for any $x \in \ell^p$ write $x = \sum_{n=1}^{\infty} x_n e_n$ (convergence in ℓ^p). By the continuity and linearity of f,

$$f(x) = \sum_{n=1}^{\infty} x_n f(e_n) = \sum_{n=1}^{\infty} x_n y_n = \Phi(y)(x).$$

Thus $f = \Phi(y)$, and Φ is onto.

Since Φ is a linear isometry and surjective, it is an isometric isomorphism. Therefore $(\ell^p)^* \cong \ell^q$. □

Remark 2.1. The proposition fails for $p = 1$ and $p = \infty$. Instead, one has $(\ell^1)^* \cong \ell^\infty$, while $(\ell^\infty)^*$ is strictly larger than ℓ^1 (it can be identified with the space of finitely additive measures)

Theorem 2.1 (Jordan–von Neumann Theorem)**.** Let $(X, \|\cdot\|)$ be a normed space over $\mathbb{K}$ ($\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$). Then the norm $\|\cdot\|$ is induced by an inner product if and only if it satisfies the **parallelogram law**:

$$\|x+y\|^2 + \|x-y\|^2 = 2\left(\|x\|^2 + \|y\|^2\right), \qquad \forall x, y \in X.$$

Moreover, if this holds, the inner product is uniquely determined by the norm through the **polarization identities**.

We prove the theorem separately for real and complex cases.

Proposition 2.2 (Real case)**.** Let $(X, \|\cdot\|)$ be a real normed space whose norm satisfies the parallelogram law. Define a mapping $\langle\cdot,\cdot\rangle : X \times X \to \mathbb{R}$ by

$$\langle x, y\rangle = \frac{1}{4}\left(\|x+y\|^2 - \|x-y\|^2\right), \qquad x, y \in X.$$

Then $\langle\cdot,\cdot\rangle$ is an inner product on X and $\|x\| = \sqrt{\langle x, x\rangle}$ for all $x \in X$.

Proof. We verify the axioms of an inner product step by step.
1. Symmetry: Clearly $\langle x, y\rangle = \langle y, x\rangle$ from the definition.
2. Positivity: $\langle x, x\rangle = \frac{1}{4}\|2x\|^2 = \|x\|^2 \geq 0$, and $\langle x, x\rangle = 0$ iff $\|x\| = 0$ iff $x = 0$.
3. Additivity in the first argument (the hardest part). We first show that $\langle x+z, y\rangle = \langle x, y\rangle + \langle z, y\rangle$ for all $x, y, z \in X$.

Define the function $f(x, y) = \|x+y\|^2 - \|x-y\|^2$. Then $\langle x, y\rangle = \frac{1}{4}f(x, y)$. Using the parallelogram law repeatedly:

$$\begin{aligned} f(x+z, y) + f(x-z, y) &= \|x+z+y\|^2 - \|x+z-y\|^2 + \|x-z+y\|^2 - \|x-z-y\|^2 \\ &= \|x+z+y\|^2 + \|x-z+y\|^2 - \left(\|x+z-y\|^2 + \|x-z-y\|^2\right). \end{aligned}$$

Apply the parallelogram law to the pairs $(x+y, z)$ and $(x-y, z)$:

$$\begin{aligned} \|x+z+y\|^2 + \|x-z+y\|^2 &= 2\|x+y\|^2 + 2\|z\|^2, \\ \|x+z-y\|^2 + \|x-z-y\|^2 &= 2\|x-y\|^2 + 2\|z\|^2. \end{aligned}$$

Thus

$$f(x+z, y) + f(x-z, y) = 2\|x+y\|^2 + 2\|z\|^2 - 2\|x-y\|^2 - 2\|z\|^2 = 2f(x, y).$$

Hence

$$\langle x+z, y\rangle + \langle x-z, y\rangle = 2\langle x, y\rangle. \tag{1}$$

Now, replacing z by x in (1) gives

$$\langle 2x, y\rangle + \langle 0, y\rangle = 2\langle x, y\rangle.$$

But $f(0, y) = \|y\|^2 - \|-y\|^2 = 0$, so $\langle 0, y\rangle = 0$. Thus $\langle 2x, y\rangle = 2\langle x, y\rangle$.

Next, set $u = x+z$ and $v = x-z$. Then $x = \frac{u+v}{2}$ and $z = \frac{u-v}{2}$. From (1) we have

$$\langle u, y\rangle + \langle v, y\rangle = 2\langle \tfrac{u+v}{2}, y\rangle = \langle u+v, y\rangle.$$

Thus $\langle u+v, y\rangle = \langle u, y\rangle + \langle v, y\rangle$, which is additivity.
4. Homogeneity in the first argument. From additivity we get $\langle nx, y\rangle = n\langle x, y\rangle$ for integers n. For rational $r = m/n$, we have $\langle rx, y\rangle = r\langle x, y\rangle$. By continuity of the norm (and hence of $\langle\cdot,\cdot\rangle$), homogeneity extends to all real scalars.

Thus $\langle\cdot,\cdot\rangle$ is a real inner product, and clearly $\|x\|^2 = \langle x, x\rangle$. □

Proposition 2.3 (Complex case)**.** Let $(X, \|\cdot\|)$ be a complex normed space whose norm satisfies the parallelogram law. Define a mapping $\langle\cdot,\cdot\rangle : X \times X \to \mathbb{C}$ by

$$\langle x, y\rangle = \frac{1}{4}\sum_{k=0}^{3} i^k \|x + i^k y\|^2,$$

where i is the imaginary unit. Then $\langle\cdot,\cdot\rangle$ is an inner product on X and $\|x\| = \sqrt{\langle x, x\rangle}$ for all $x \in X$.

Proof. The proof follows a similar pattern but is more involved due to the sesquilinearity.
1. Conjugate symmetry: We show $\langle y, x\rangle = \overline{\langle x, y\rangle}$. Observe that for any $z \in X$, $\|z\|^2 = \|\overline{i^k} z\|^2$ because scaling by a complex number of modulus 1 does not change the norm. Specifically, $\|i^k z\| = \|z\|$. Then

$$\begin{aligned}\overline{\langle x, y\rangle} &= \frac{1}{4}\sum_{k=0}^{3}(-i)^k\|x + i^k y\|^2 = \frac{1}{4}\sum_{k=0}^{3}(-i)^k\|(-i)^k x + y\|^2 \\ &= \frac{1}{4}\sum_{k=0}^{3} i^{-k}\|y + i^{-k} x\|^2 = \frac{1}{4}\sum_{k=0}^{3} i^k\|y + i^k x\|^2 = \langle y, x\rangle,\end{aligned}$$

after reindexing the sum (let $k' = -k \mod 4$).
2. Positivity: $\langle x, x\rangle = \frac{1}{4}\sum_{k=0}^{3} i^k\|x + i^k x\|^2$. Now $\|x + i^k x\|^2 = |1 + i^k|^2\|x\|^2$. Since $|1 + i^k|^2 = 2$ for $k = 0, 2$ and 0 for $k = 1, 3$, we get

$$\langle x, x\rangle = \frac{1}{4}\left(1 \cdot 2\|x\|^2 + i^2 \cdot 2\|x\|^2\right) = \frac{1}{4}(2\|x\|^2 - 2\|x\|^2) = \|x\|^2.$$

Thus $\langle x, x\rangle = \|x\|^2 \geq 0$, and equals 0 only when $x = 0$.
3. Linearity in the first argument: The proof of additivity follows the same idea as in the real case, but one must work with the four-term polarization identity. One first proves an analogue of (1) for the complex polarization form, then deduces additivity. Homogeneity for real scalars follows as before; for complex scalars, one first proves $\langle ix, y\rangle = i\langle x, y\rangle$ by direct computation using the definition, then combines with real homogeneity to get full complex homogeneity.
4. Sesquilinearity: From the above, $\langle\cdot,\cdot\rangle$ is linear in the first argument and conjugate symmetric, hence conjugate linear in the second argument.

Therefore $\langle\cdot,\cdot\rangle$ is a complex inner product inducing the given norm. □

Remark 2.2. The converse of the theorem is straightforward: if the norm comes from an inner product, then the parallelogram law holds, as shown earlier.

3 Application of B.L.T. Theorem: Weak Derivatives

Definition 3.1 (Weak Derivative)**.** For a function $u \in L^2(\Omega)$ and a multi-index α, we say that $v \in L^2(\Omega)$ is the α-th weak derivative of u if for every test function $\phi \in C_c^\infty(\Omega)$,

$$\int_\Omega u(x) D^\alpha \phi(x)\, dx = (-1)^{|\alpha|} \int_\Omega v(x)\phi(x)\, dx.$$

We denote $v = D^\alpha u$.

Note that for $u \in C_c^\infty(\Omega)$, the weak derivative coincides with the classical derivative. Now, for each fixed $\phi \in C_c^\infty(\Omega)$, consider the linear functional $T_\phi : C_c^\infty(\Omega) \to \mathbb{R}$ defined by

$$T_\phi(u) = \int_\Omega u(x) D^\alpha \phi(x)\, dx.$$

This functional is continuous with respect to the L^2 norm because by the Cauchy-Schwarz inequality,

$$|T_\phi(u)| \leq \|D^\alpha \phi\|_{L^2} \|u\|_{L^2}.$$

Thus, T_ϕ is a bounded linear functional on the dense subspace $C_c^\infty(\Omega)$ of $L^2(\Omega)$. By the B.L.T. Theorem, T_ϕ can be uniquely extended to a bounded linear functional on $L^2(\Omega)$.

Now, if we want to define the weak derivative $D^\alpha u$ for $u \in L^2(\Omega)$, we note that the map $\phi \mapsto T_\phi(u)$ is a linear functional on $C_c^\infty(\Omega)$. However, it is not immediately obvious that this functional is given by integration against an L^2 function. But if it is, then that function is the weak derivative. The Riesz representation theorem for Hilbert spaces tells us that every bounded linear functional on $L^2(\Omega)$ is represented by an L^2 function. Therefore, the extension of T_ϕ to $L^2(\Omega)$ is given by

$$T_\phi(u) = \int_\Omega v(x)\phi(x)\, dx$$

for some $v \in L^2(\Omega)$. Then we set $D^\alpha u = v$.

Thus, the B.L.T. Theorem allows us to extend the notion of derivative from smooth functions to L^2 functions in a weak sense.

The above illustration is completed by Deepseek. However, it is not correct as it stands. The key issue is that not every L^2 function has an L^2 weak derivative. The key point is that we cannot guarantee

3.1 The Sobolev Space $W^{1,2}(0,1)$

Definition 3.2 (Sobolev space $W^{1,2}(0,1)$)**.** The Sobolev space $W^{1,2}(0,1)$ is defined as

$$W^{1,2}(0,1) = \left\{ f \in L^2(0,1) : \exists g \in L^2(0,1) \text{ such that } \int_0^1 f\phi' = -\int_0^1 g\phi,\ \forall \phi \in C_c^\infty(0,1) \right\}.$$

The function g (unique by density) is called the **weak derivative** of f and denoted f'.

Proposition 3.1 (Alternative characterization)**.** The space $C^1[0,1]$ is dense in $W^{1,2}(0,1)$ with respect to the norm

$$\|f\|_{W^{1,2}} = \left(\|f\|_{L^2}^2 + \|f'\|_{L^2}^2\right)^{1/2},$$

where f' denotes the weak derivative for $f \in W^{1,2}$ and the classical derivative for $f \in C^1$.

3.2 Constructing the Weak Derivative Operator via B.L.T.

Theorem 3.1 (Extension of the classical derivative)**.** Consider the classical derivative operator $D : C^1[0,1] \to L^2(0,1)$ defined by $Df = f'$ (classical derivative). Then D extends uniquely to a bounded linear operator $\tilde{D} : W^{1,2}(0,1) \to L^2(0,1)$ satisfying $\|\tilde{D}\| \leq 1$, and this extension coincides with the weak derivative.

Proof. **Step 1: D is bounded on** $(C^1, \|\cdot\|_{W^{1,2}})$. For $f \in C^1[0,1]$,

$$\|Df\|_{L^2} = \|f'\|_{L^2} \leq \left(\|f\|_{L^2}^2 + \|f'\|_{L^2}^2\right)^{1/2} = \|f\|_{W^{1,2}}.$$

Thus $D : (C^1, \|\cdot\|_{W^{1,2}}) \to L^2(0,1)$ is bounded with $\|D\| \leq 1$.

Step 2: Apply B.L.T. theorem. Since $C^1[0,1]$ is dense in $W^{1,2}(0,1)$ with respect to $\|\cdot\|_{W^{1,2}}$, the B.L.T. theorem guarantees a unique bounded extension $\tilde{D} : W^{1,2}(0,1) \to L^2(0,1)$ with $\|\tilde{D}\| \leq 1$.

Step 3: $\tilde{D}$ coincides with the weak derivative. Let $f \in W^{1,2}(0,1)$ and choose $f_n \in C^1[0,1]$ such that $f_n \to f$ in $W^{1,2}$. Then $\tilde{D}f = \lim_{n\to\infty} Df_n = \lim_{n\to\infty} f_n'$ in L^2. For any test function $\phi \in C_c^\infty(0,1)$, we have

$$\int_0^1 \tilde{D}f \cdot \phi = \lim_{n\to\infty} \int_0^1 f_n'\phi = -\lim_{n\to\infty} \int_0^1 f_n\phi' = -\int_0^1 f\phi'.$$

Thus $\tilde{D}f$ satisfies the definition of the weak derivative of f. □

Remark 3.1. The key points:

1. We do **not** claim that every L^2 function has an L^2 weak derivative.
2. Instead, we start with C^1 functions equipped with the $W^{1,2}$ norm, which is strictly stronger than the L^2 norm.

3. The completion of C^1 under this norm is exactly $W^{1,2}$, not all of L^2.
4. The extended operator $\tilde{D}$ is defined only on $W^{1,2}$, not on all of L^2.

3.3 A Concrete Counterexample

Proposition 3.2 (Not every L^2 function has an L^2 weak derivative)**.** The function $f(x) = |x - 1/2|^{1/2}$ belongs to $L^2(0,1)$ but not to $W^{1,2}(0,1)$.

Proof. First, compute

$$\int_0^1 |f(x)|^2 dx = \int_0^1 |x - 1/2| dx = \frac{1}{4} < \infty,$$

so $f \in L^2(0,1)$.

Suppose f had a weak derivative $g \in L^2(0,1)$. Then for all $\phi \in C_c^\infty(0,1)$,

$$\int_0^1 g\phi = -\int_0^1 f\phi' = -\int_0^1 |x - 1/2|^{1/2}\phi'(x)dx.$$

Take ϕ such that $\phi(x) = 1$ near $1/2$. The right-hand side is essentially

$$-\int |x - 1/2|^{1/2}\phi'(x)dx,$$

which diverges because $|x-1/2|^{1/2}$ is not absolutely continuous and its classical derivative $\frac{1}{2}|x - 1/2|^{-1/2}\operatorname{sgn}(x - 1/2)$ is not in L^2 (its square integral diverges logarithmically at $1/2$). More rigorously, one can show that no $g \in L^2$ can satisfy the integration-by-parts formula. □

4 Application of the B.L.T. Theorem: Fourier Transform on $L^2(\mathbb{R})$

Definition 4.1. The Schwartz space $\mathcal{S}(\mathbb{R})$ is the space of all infinitely differentiable functions $f : \mathbb{R} \to \mathbb{C}$ such that for all nonnegative integers n, m,

$$\sup_{x\in\mathbb{R}} |x^n f^{(m)}(x)| < \infty.$$

Definition 4.2. For $f \in \mathcal{S}(\mathbb{R})$, the Fourier transform $\mathcal{F}(f)$ is defined by

$$\mathcal{F}(f)(\xi) = \hat{f}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-2\pi i x\xi}\, dx, \quad \xi \in \mathbb{R}.$$

Proposition 4.1 (Plancherel's Theorem on $\mathcal{S}(\mathbb{R})$)**.** For $f \in \mathcal{S}(\mathbb{R})$, we have

$$\|\hat{f}\|_{L^2} = \|f\|_{L^2}.$$

Moreover, the Fourier transform $\mathcal{F} : \mathcal{S}(\mathbb{R}) \to \mathcal{S}(\mathbb{R})$ is a bijection.

Now note that $\mathcal{S}(\mathbb{R})$ is dense in $L^2(\mathbb{R})$. The Fourier transform $\mathcal{F}$ on $\mathcal{S}(\mathbb{R})$ is a linear operator and, by Plancherel's theorem, it is an isometry (hence bounded) with respect to the L^2 norm.

Theorem 4.1 (B.L.T. Theorem)**.** Let X and Y be normed linear spaces, and let X_0 be a dense subspace of X. If $T_0 : X_0 \to Y$ is a bounded linear operator, then there exists a unique bounded linear operator $T : X \to Y$ extending T_0, and $\|T\| = \|T_0\|$.

Applying the B.L.T. Theorem with $X = Y = L^2(\mathbb{R})$, $X_0 = \mathcal{S}(\mathbb{R})$, and $T_0 = \mathcal{F}$, we obtain a unique bounded linear operator (which we still denote by $\mathcal{F}$) from $L^2(\mathbb{R})$ to $L^2(\mathbb{R})$ such that:

- $\mathcal{F}(f) = \hat{f}$ for all $f \in \mathcal{S}(\mathbb{R})$.
- $\|\mathcal{F}(f)\|_{L^2} = \|f\|_{L^2}$ for all $f \in L^2(\mathbb{R})$ (by continuity and density).

This extension is called the Fourier transform on $L^2(\mathbb{R})$. Moreover, since the Fourier transform on $\mathcal{S}(\mathbb{R})$ is invertible and the inverse is also an isometry, we can similarly extend the inverse Fourier transform to $L^2(\mathbb{R})$, and it will be the inverse of the extended Fourier transform. Thus, the Fourier transform on $L^2(\mathbb{R})$ is a unitary operator.